1.	***2.***	***3.***	***4.***	***5.***	***6.***	***7.***	***8.***	***9.***	***10.***
A	***B***	***D***	***C***	***C***	***B***	***C***	***A***	***D***	***C***
11.	***12.***	***13.***	***14.***	***15.***	***16.***	***17.***	***18.***	***19.***	***20.***
B	***B***	***A***	***B***	***A***	***D***	***D***	***B***	***C***	***A***

1 Methyl mercaptan, CH_3SH, is a substance often used to impart a smell to natural gas in a pipeline.
The chemical equation for the combustion of CH_3SH is as shown below.

$$CH_3SH + 3O_2 \rightarrow CO_2 + SO_2 + 2H_2Oa$$

A 10 cm^3 sample of CH_3SH is exploded with 60 cm^3 of oxygen & the resultant gas mixture is passed into excess NaOH(aq).
What is the percentage of the volume of the resultant gas mixture dissolved in NaOH? [All gas volumes are measured at room temperature & pressure.]

A √ 40.0% **B** 28.5% **C** 20.0% **D** 14.3%

	CH_3SH +	$3O_2 \rightarrow$	CO_2 +	SO_2 + $2H_2O$
$V_{initial}$ / cm^3	10	60	0	0
V_{final} / cm^3	0	30	10	10

Both acidic CO_2 + SO_2 are absorbed by NaOH(aq)

$\therefore$ % vol decrease = $\frac{20}{50} x\ 100$ = 40.0%

Ans: **A**

2 Calcium carbide, CaC_2, can be manufactured from calcium oxide & coke, C, according to the following equation.

$$CaO(s) + 3C(s) \rightarrow CaC_2(s) + CO(g)$$

What is the maximum mass of calcium carbide, in kg, that can be obtained from 40.0 kg of calcium oxide & 40.0 kg of coke?

A 40.0 **B** √ 45.7 **C** 80.0 **D** 214

	CaO(s) +	3C(s) $\rightarrow$	$CaC_2(s)$ + CO(g)
Amount / mol	$\frac{40}{56}$ = 0.714	$\frac{40}{12}$ = 3.33	

$\Rightarrow$ CaO is limiting. $\therefore$ max mass of $CaC_2(s)$ obtainable = $\frac{40}{56} x\ 64.0$ = 45.7 g

Ans: **B**

3 *Use of the Data Booklet is relevant to this question.*
Which underlined element in the species has an oxidation number of +6?

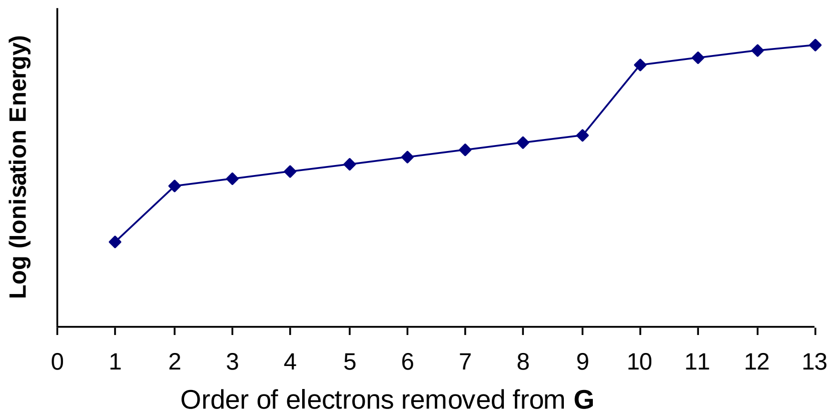
A $\underline{Sb}Cl_5^{2-}$ **B** $\underline{Pt}Cl_6^{2-}$ **C** $H_2\underline{P}_2O_7^{2-}$ **D** √ $\underline{Re}OCl_4$

O.N.: **+3** **+4** **+5** **+6**

(O.N. of C*l* = –1; O.N. of H = +1; O.N. of O = –2)

Ans: **D**

4 **G** is an element in period 4 of the Periodic Table. The first 13 ionisation energies of **G** are plotted against the order of the removal of electrons as shown below.



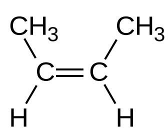
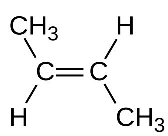
What is the formula of the compound formed when **G** reacts with oxygen?

A	**G**O	**B**	**G**O_2	**C** √	**G**$_2$O	**D**	**G**$_2O_3$

1st sharp increase occurs between 1st & 2nd IE's ⇒ **G** is in Group **1** (has 1 valence e per atom)

Ans: **C**

5 In which pair of compounds is the first member less volatile (has higher bp) than the second one?

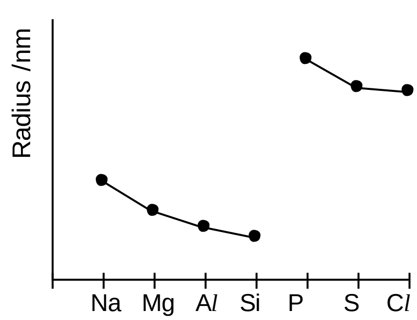
A	CO_2	CS_2	Both are non-polar & linear in shape with weak intermolecular td-id interactions. CO_2 has a smaller e cloud & thus weaker td-id interactions; hence a lower bp.
B	PH_3	NH_3	PH_3 is non-polar with weaker intermolecular td-id interactions while NH_3 is polar with stronger intermolecular H bonding. Thus PH_3 has a lower bp
C √			*Cis*-but-2-ene is **slightly polar** while trans-but-2-ene is non-polar. Thus, *cis*-but-2-ene has slightly stronger intermolecular forces of attraction & a higher bp.
D	$(CH_3)_3CH$	$CH_3CH_2CH_2CH_3$	Both are **non-polar**; $(CH_3)_3CH$ is branched with a smaller surface area of contact between molecules for td-id interactions & thus has a lower bp.

Ans: **C**

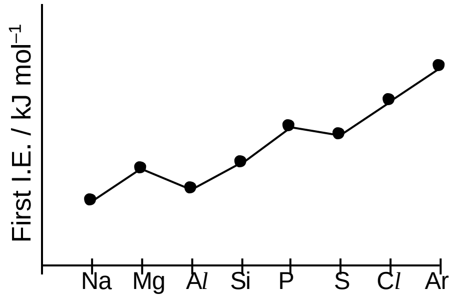
6 Which sketches show the correct trend in the stated property for the elements in period 3 of the Periodic Table?

1 **atomic radius**

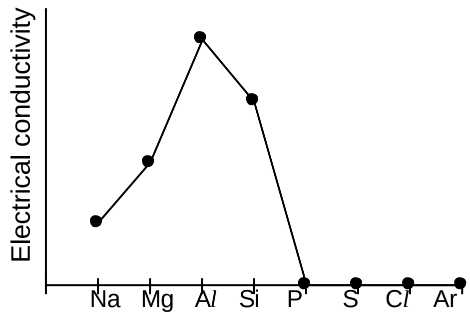
This graph shows the trend of the ionic radius instead. Atomic radius decreases across a period due to increase nuclear charges, but insignificant increase in shielding effect.



2 √ **first ionisation energy** This graph shows the correct trend of 1st IE's of period 3 elements.



3 **electrical conductivity** Si has should have a **lower** electrical conductivity than Na, Mg & A*l*.



A **1** only **B** √ **2** only **C** **1** & **2** only **D** **2** & **3** only

Ans: **B**

7 The shapes of three species **P, Q,** & **R** are respectively bent, square planar & trigonal pyrimidal.
Which of the following can be **P, Q** & **R**?

Candidates need only to check on those familiar ones to verify that they do not have the stated shape to rule out the choice. E.g. F_2O is similar to H_2O**;** CS_2 is similar to CO_2.

	P	**Q**	**R**
A	F_2O (2 lp + 2 bp; bent)	ICl_4^- (2 lp + 4 bp; sq planar)	BCl_3 (3 bp only; trig planar)
B	CS_2 (2 bp only; linear)	BH_4^- (4 bp only; Td)	ICl_3 (2 lp + 3 bp; T-shape)
C √	$SnCl_2$ (1lp + 2 bp; bent)	BrF_4^- (2 lp + 4 bp; sq planar)	SbF_3 (1 lp + 3 bp; trig pyramidal)
D	ICl_2^- (3 lp + 2 bp; linear)	XeF_4 (2 lp + 4 bp; sq planar)	SO_3^{2-} (1 lp + 3 bp; trig pyramidal)

Ans: **C**

8 Which sequence correctly shows the arrangement of bonds in increasing polarity?

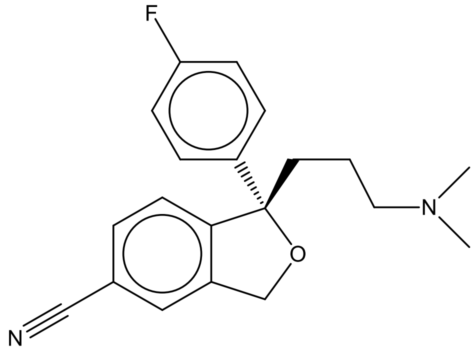
Note: the greater the difference in the electronegativity of the atoms, the more polar is the bond.

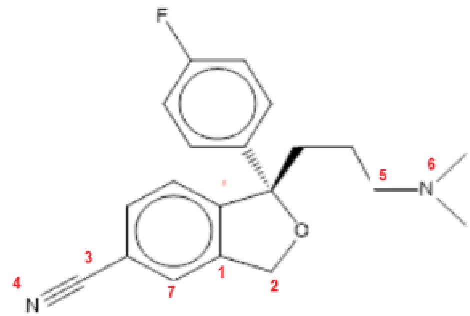
A √ O–O < O–C*l* < O–S **B** O–O < O–S < O–C*l*

C O–S < O–C*l* < O–O **D** O–S < O–O < O–C*l*

Ans: **A**

9 Which bond is **not** present in the following molecule?





A a σ bond formed by sp^3–sp^2 orbital overlap between two C atoms (e.g., between C1 & C2)

B a π bond formed by p–p orbital overlap between C & N atoms (between C3 & N4)

C a σ bond formed by sp^2–sp^2 orbital overlap between two C atoms (e.g., between C1 & C7)

D √ a σ bond formed by sp^3–sp^2 orbital overlap between C & N atoms (between C5 & N6 is a σ bond formed by **sp^3–sp^3** orbital overlap)

Ans: **D**

10 Given: $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$ $\Delta H = -114\ kJ\ mol^{-1}$;
what is the most likely ΔH value for $Ba(OH)_2(aq) + 2HCl(aq) \rightarrow BaCl_2(aq) + 2H_2O(l)$?

A $-57\ kJ\ mol^{-1}$ **B** $-76\ kJ\ mol^{-1}$ **C** √ $-114\ kJ\ mol^{-1}$ **D** $-228\ kJ\ mol^{-1}$

From info given: when 2 mol of $H_2O(l)$ are formed from a strong acid-strong base reaction; –114 kJ are given off.

As $Ba(OH)_2(aq) + 2HCl(aq) \rightarrow BaCl_2(aq) + 2H_2O(l)$ is also a strong acid-strong base reaction; same amount of heat should be released.

Ans: **C**

11 The table below shows the standard enthalpy change of hydrogenation to form cyclohexane of three compounds.

compound	$\Delta H^\ominus_{hydrogenation}$ / $kJ\ mol^{-1}$
benzene	–206
cyclohexa-1,3,5-triene	–360
cyclohexene	–120

Which statement is correct?

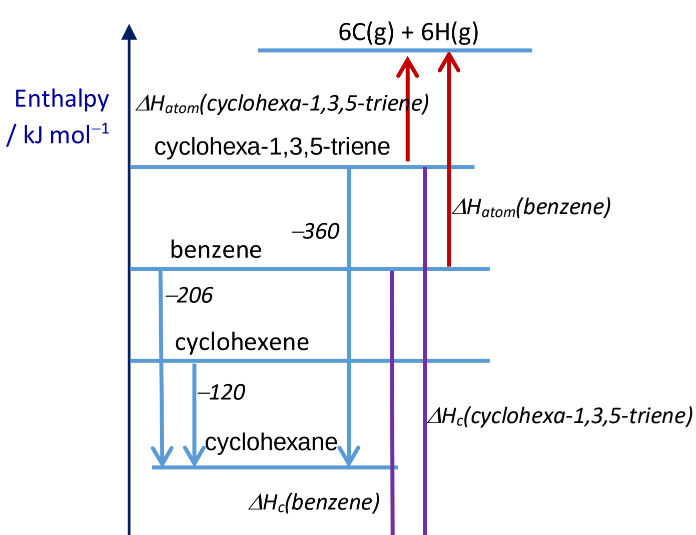
A Cyclohexa-1,3,5-triene is more stable than benzene. ×

B √ Combustion of benzene is less exothermic than cyclohexa-1,3,5-triene.

C The C=C bond in cyclohexene is weaker than those in cyclohexa-1,3,5-triene. × (should be the same)

D The enthalpy change of atomisation of benzene is smaller than that of cyclohexa-1,3,5-triene. ×

Refer to the following diagram, answer is **B**.



12 Which molecules do **not** have all the carbon atoms lying in one plane?

1 √ benzene **2** × cyclohexene **3** √ ethene **4** × ethylbenzene

A **2** only **B** √ **2** & **4** only **C** **1**, **3** & **4** **D** **2**, **3** & **4**

All C atoms in benzene & ethene are *sp*^{2} hybridised & hence lie in 1 plane (trigonal planar); while cyclohexene & ethylbenzene have *sp*^{3} hybridised C atoms (tetrahedral arrangement) & cannot have all C atoms lying in a plane.

Ans: **B**

13 A catalytic converter is part of the exhaust system of many modern cars.
Which reactions occur in a catalytic converter?

1 √ $2CO + 2NO \rightarrow 2CO_2 + N_2$

2 × $CO_2 + NO \rightarrow CO + NO_2$

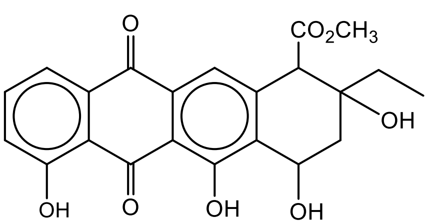
3 × $2SO_2 + 2NO \rightarrow 2SO_3 + N_2$

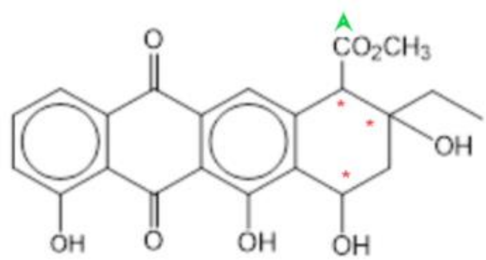
A √ **1** only **B** **1** & **2** only **C** **1** & **3** **D** **1**, **2** & **3**

Catalytic converters are to convert harmful to less harmful gases before they leave the exhaust pipes. As CO is very toxic & SO_3 is responsible for acid rain, they should not be formed.

Ans: **A**

14 Aklavinone shown below is a tetracycline antibiotic.





Which combination of the number of chiral centres & of the number of sp^2 & sp^3 hybridised carbon atoms does it possess?

	number of chiral centres	*number of sp^2 hybridised C*	*number of sp^3 hybridised C*
A	3	14	8
B √	3 (C with *)	15	7
C	5	14	8
D	5	15	7

The compound has 15 *sp*^{2} hybridised C: all C's of benzene rings & those with C=O bonds (including the 1 with green arrow).

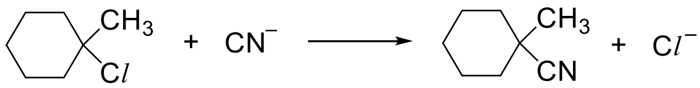
Ans: **B**

15 Species with the molecular formula CH_3 can act as an electrophile, a free radical or a nucleophile depending on the number of valence electrons on the central carbon atom.
How many valence electrons must be present for CH_3 to act in the different ways?

	as an electrophile	*as a free radical*	*as a nucleophile*
A √	6 (CH_3^+)	7 ($\bullet CH_3$)	8 ($:CH_3^-$)
B	6	8	7
C	7	6	8
D	8	7	6

Ans: **A**

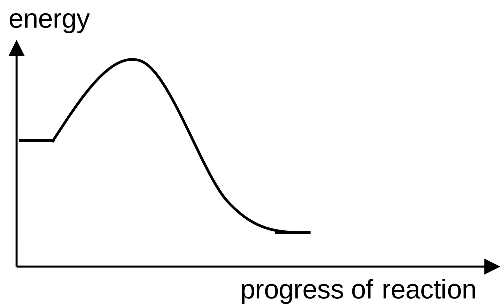
16 The reaction between 1-chloro-1-methylcyclohexane & ethanolic potassium cyanide is shown below:



Which statements regarding the reaction are likely to be true?

1 √ The overall order of reaction is 1. (3° RX undergoes S_N1 mechanism involving 2 steps; slow step is RX → $R^+ + X^-$)

2 × The energy profile diagram for the reaction is as shown below: (S_N1 (2-steps) mechanism has 2 humps with 2 different E_a's)



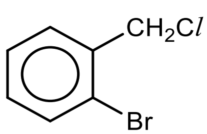
3 × 1-Bromo-1-methylcyclohexane will react through the same mechanism, but less readily. (C-Br bond is weaker & should be more readily broken.)

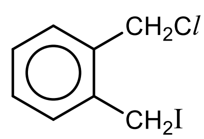
A **1**, **2** & **3** **B** **2** & **3** only **C** **1** & **2** only **D** √ **1** only

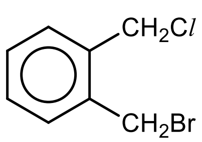
Ans: **D**

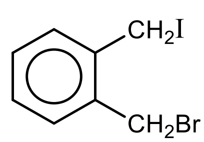
17 Equal amounts of each compound were heated with excess NaOH(aq). Excess HNO_3(aq) & $AgNO_3$(aq) were then added & the precipitate collected & dried.

Which compound will produce the largest mass of precipitate?

A 

B 

C 

D √ 

Assume 1 mole of each compound is used & reaction is complete.

A: gives 1 mol Ag*Cl*(s) (143.4 g)

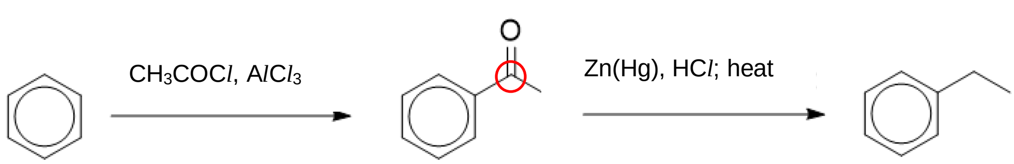
B: gives 1 mol Ag*Cl*(s) + 1 mol AgI(s) (total = 378.2 g)

C: gives 1 mol Ag*Cl*(s) + 1 mol AgBr(s) (total = 331.2 g)

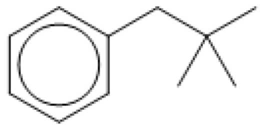
D: gives 1 mol AgI(s) + 1 mol AgBr(s) (total = 422.6 g)

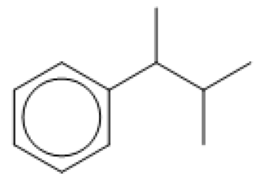
Ans: **D**

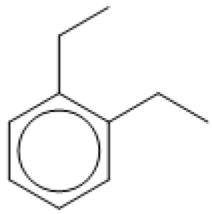
18 Alkylbenzenes can be synthesised from benzene by introducing an acyl group via electrophilic substitution, followed by the reduction of the carbonyl group with zinc amalgam & hydrochloric acid. For example, ethylbenzene can be synthesised by the following route.

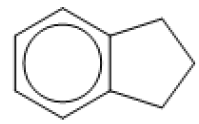


Which compound **cannot** be synthesised through the same route?

A 

B √ 

C 

D 

From the info given, benzylic C (circled in red) has a double bond & thus cannot be attached to an R group.
Ans: **B**

19 Poly(tetrafluoroethene) is a polymer used for coating non-stick kitchen utensils & replacing bone joints. One of the stages in the manufacture of the polymer is as below.

$$2CHClF_2(g) \rightleftharpoons C_2F_4(g) + 2HCl(g) \qquad \Delta H = +128 \text{ kJ mol}^{-1}$$

Which changes will increase the amount of tetrafluoroethene, C_2F_4, present at equilibrium?

1 decrease pressure √

2 increase temperature √

3 add a catalyst ×

4 add more $CHClF_2(g)$ √

A **1** & **2** **B** **1** & **3** **C** √ **1**, **2** & **4** **D** **2**, **3** & **4**

1 is true - according to the LCP, ↓ P causes eqm to shift to the side with more gaseous components, i.e., forward direction, amount of C_2F_4 ↑.
2 is true - according to the LCP, ↑ T causes eqm to shift to forward endothermic reaction to reduce the E in the system; amount of C_2F_4 ↑.
4 is true – adding more reactant causes eqm to shift to the right to form more C_2F_4.
3 is false - adding a catalyst has no effect on the eqm position; thus no change in amount of C_2F_4.
Ans: **C**

20 A pure sample of dinitrogen pentoxide, N_2O_5, is introduced into an evacuated vessel & heated to a constant temperature such that the equilibrium below is established.

$$2N_2O_5(g) \rightleftharpoons 4NO_2(g) + O_2(g)$$

The value of the pressure is then found to be 50% greater than if only $N_2O_5(g)$ were present.
What is the mole fraction, *x*, of oxygen in this equilibrium?

A √ 0.11 **B** 0.20 **C** 0.33 **D** 0.50

Let initial $\eta_{N2O5(g)}$ be 2 mol

	$2N_2O_5(g)$ ⇌	$4NO_2(g)$ +	$O_2(g)$	
Initial $\eta_{N2O5(g)}$ / mol	2	0	0	
Change / mol	–2a	+4a	+a	
Eqm $\eta_{N2O5(g)}$ / mol	2 – 2a	4a	a	η_{total} = 2 + 3a mol

Thus, 2 mol of gas exerts a pressure of p; (2 + 3a) mol of gases exert a pressure of $\frac{3}{2}$ p (50% greater);

i.e, $\frac{\frac{3}{2}p}{p} = \frac{2+3a}{2}$

On solving, $a = \frac{1}{3}$; $\eta_{total} = 2 + 3(\frac{1}{3}) = 3$ mol

mole fraction of oxygen = $\frac{\eta_{O2}}{\eta_{total}} = \frac{\frac{1}{3}}{3} = = \frac{1}{9} = 0.11$

Ans: **A**

end of paper